This is a pre-copyedited, author-produced version of an article accepted for publication in *The Philosophical Quarterly* following peer review. The version of record (Frankish, K. (2009). [Review of the book *The Architecture of the Mind*, by P. Carruthers]. *The Philosophical Quarterly*, 59(235), 371–375) is available online at: https://doi.org/10.1111/j.1467-9213.2009.618_6.x. Version of 11 October 2026.

Review of *The Architecture of the Mind* by Peter Carruthers

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(Oxford UP, 2006. Pp. xviii + 462.)

Peter Carruthers's *The Architecture of the Mind* argues for, elaborates, and defends a version of the massive modularity hypothesis, according to which the human mind is composed of a great many functionally specialized components, or modules.

The book consists of seven main chapters, plus a short conclusion. The first three chapters make the case for massive modularity. Chapter 1 begins by isolating the notion of modularity appropriate to the thesis that central cognition is modular, distinguishing it from that developed by Fodor for input and output systems. Central ('conceptual') modules, Carruthers argues, are function-specific automatic processing systems, which are partly independent of one another, and whose internal operations are largely inaccessible to the rest of cognition. The chapter then sets out three arguments for the view that the human mind is massively modular in this sense. These are: from design (complex evolved systems are typically built out of distinct sub-components); from animals (it is very likely that animal minds are massively modular, and it would be extraordinary if this architecture had not been inherited by hominids); and from computational tractability (mental processes are computational, and in order to be tractable, the computations involved must be performed by subsystems that are computationally frugal, though not necessarily *encapsulated* in the traditional sense).

The second chapter develops the case for a massively modular view of animal minds, thus providing support for the main premise of the argument from animals. Carruthers begins by arguing for the prevalence of a perception/belief/desire/planning architecture within the animal kingdom (including even insects). He then argues for the existence of distinct but interconnected modules that help to realize this basic architecture, including dual visual systems, suites of belief-generating and desire-generating modules, multiple memory systems, and suites of action-selection modules. The final section, which is more speculative, considers mental rehearsal, in which activated motor schemata generate perceptual feed-

back that is then globally broadcast to all belief-forming and motivational subsystems. By this means, Carruthers suggests, a creature would be able to calculate some of the consequences of potential actions, thereby vastly extending its problem-solving abilities. Carruthers argues that the capacity for such mental rehearsal is present in at least some primates and that it was possessed by early species of hominid.

Chapter 3 considers some of the distinctively human modules that have been added in the transition from our great-ape ancestors to ourselves. The first section rebuts alternative views, on which the transition involved just one major adaptation or simply a growth in brain size. Carruthers then proceeds to argue for the existence of various human-specific modules, both cognitive and motivational, embedded within the pre-existing architecture. These include modules for simple physics, biological classification, mind-reading, language, imitation, mating, status, kin selection, and dealing with social norms. The likely patterns of connectivity between these modules are also discussed.

Chapters 4–7 address challenges to massive modularity. The core challenge is to explain how a massively modular mind could display the flexibility and creativity characteristic of human cognition. Chapter 4 focuses on flexibility, under three aspects: stimulus independence (the ability to entertain thoughts that are independent of current stimuli), flexibility of content (the ability to frame thoughts that combine the outputs of different modules), and flexibility of reasoning and decision-making (the ability to acquire new reasoning strategies and patterns of thinking). In each case, Carruthers proposes that natural language has a crucial role to play. Stimulus-independence, he suggests, has two sources: mental rehearsal of the sort discussed earlier, massively boosted by the development of language, and episodic memory, which evokes imagery that can set off chains of memories unrelated to present stimuli. Content integration, he argues, is effected by the language-production system, with integrated thoughts taking the form of Mentalese representations of the phonology of natural language sentences. Flexibility of reasoning, meanwhile, is a consequence of our ability to rehearse utterances in inner speech; such utterance-rehearsals constitute what is in effect a new reasoning system, whose operations are under active control and can be modified through imitation and instruction. (Here Carruthers makes a link with currently popular *dual-system* theories of reasoning, of the sort advocated by Jonathan Evans, Keith Stanovich, and others.)

Chapter 5 considers creativity. Carruthers begins by establishing some constraints on theorizing about human creativity. Crucial among these is the claim that child pretend play and adult creativity have the same cognitive basis, which resides in a capacity to reason with novel suppositions. Carruthers then proceeds to develop what he calls the 'creative-action theory of creativity', according to which creative supposition is grounded in an ability to generate novel action schemata, which can then be mentally rehearsed, as in the problem-solving episodes described earlier. The crucial adaptation for human creativity, Carruthers suggests, was the development of dispositions to rehearse action schemata that are only loosely connected with the present context (by 'boosting' weakly activated concepts into stronger ones). Again, Carruthers suggests that language played an important role in extending this process. By generating novel speech-act schemata in inner speech, we can engage in propless pretend play and conjure up novel thoughts with cross-modular contents. The chapter goes on to explore some of the consequences of this view, with particular emphasis on its implications for autism. Further sections contrast the proposed view with the rival thought-based view of creativity proposed by Nichols and Stich, and with the view that creativity rests on a non-linguistic domain-general capacity for structure-mapping.

Chapter 6 looks at the human capacity for science. Carruthers points out that scientific enquiry has three components: the generation of creative hypotheses, the exploration of the consequences of these hypotheses, and the selection of the best candidate through abductive reasoning (inference to the best explanation). He argues that these activities are also required for tracking prey, that they were originally deployed in that context, and that there have been cognitive adaptations to support them — specifically the development of an innate capacity for abductive inference. This view, Carruthers argues, is more plausible than the alternative, on which a capacity for scientific reasoning was selected for its role in theory formation during infancy — a view which ignores the social dimension of scientific enquiry. The second part of the chapter considers how the skills required for scientific enquiry could be supported by a modular mind, focusing in particular on the capacity for abductive inference. Carruthers argues that there is unlikely to be an abductive reasoning module and suggests that abductive reasoning piggybacks on the mechanisms involved in dealing with testimony and the interpretation of speech. Like testimony, linguistically articulated hypotheses will be globally broadcast to the various conceptual modules, generating measures of their accu-

racy, consistency, coherence, simplicity, and informativeness. In some cases simple heuristics will then directly generate a belief, while in others an episode of meta-level practical reasoning will follow, culminating in a decision to take a hypothesis as true for the purposes of reasoning and decision-making.

Chapter 7 considers some features of distinctively human practical reason, beginning with reasoning about ends. Carruthers argues that such reasoning can co-opt the rehearsal-based mechanisms for domain-general theoretical reasoning described in the previous chapters. He suggests that two further modifications were required: first, the development of a disposition to recast motivational states as theoretical beliefs to the effect that the desired outcomes would be good, and, second, the acquisition of either an innate desire to do what we judge best or a disposition to translate value judgements into judgements of obligation, which the normative reasoning module can then convert into desires. The chapter also discusses the nature of intention, doxastic theories of motivation (which Carruthers rejects), and the role of the conscious will. With regard to the last, Carruthers agrees with Wegner that the conscious will is illusory, since, on his view, conscious decisions do not terminate the decision-making process, but merely serve as inputs to subsequent non-conscious decision-making.

This is an important, timely, and enormously impressive book. The massive modularity hypothesis is central to much contemporary work in evolutionary psychology, yet it is still to win widespread acceptance and is often misunderstood and rejected out of hand, both by philosophers and by psychologists. There is thus a need for a clear, comprehensive, and well-argued defence of the hypothesis, suitable for a wide range of readers. Carruthers's book meets this need superbly, and thereby sets a benchmark for all future discussions of this topic. It is (as Carruthers notes) almost absurdly ambitious, given the range of disciplines relevant to the task, but its ambition is matched by its scholarship and argumentative skill. Carruthers assembles a wealth of evidence from across the cognitive sciences and engages in disciplined but creative abductive reasoning, conducted with exceptional clarity. Carruthers notes that some philosophers may not recognize such empirically based theorizing as a form of philosophy, but appeals to Aristotle and Hume as role models. At any rate, it is a form of theorizing that is needed if we are to integrate empirical findings from different disciplines and develop global theories of mental architecture, and Carruthers's proposals will be of immense interest to philosophers interested in the organization and powers of the human mind.

How persuasive are Carruthers's arguments? The core arguments for massive modularity are, in my opinion, very persuasive indeed. Carruthers makes a strong case for thinking that a commitment to some form of massive modularity is implicit in an evolutionary view of the mind, and the map of the modular mind he draws is a highly plausible one, supported by an array of empirical data. Some readers, it is true, will question the commitments to computationalism and belief-desire psychology that shape Carruthers's project. Such readers may still profit from the book, however. It is only when combined with massive modularity that computationalism achieves its full explanatory potential, and the book illustrates how powerful the combination can be. Besides, much of what Carruthers has to say — in particular, about the global architecture of mind and the nature of flexible, distinctively human thinking — seems compatible with non-computational views of low-level cognitive processing.

At a more detailed level, the sheer richness and ambition of the book mean that there is ample scope for debate and disagreement. I shall mention one problem, with Carruthers's account of content-flexibility. According to this, the language system serves as a content-integrator, combining outputs of different modules to create cross-modular thoughts in the form of imagistic representations of natural language sentences. These are then globally broadcast, and their contents extracted by the language comprehension system and made available to the suite of modular reasoning systems. The problem is to see what a modular mind could do with a representation of a *cross-modular* content. How could it be stored as a belief by a modular memory system, or taken as an input by a modular reasoning system? Carruthers suggests that cross-modular contents could be stored in the form of quasi-perceptual mental models, but not all contents could be encoded in this way, and it is hard to see how such models could be employed in reasoning, in conjunction with other beliefs and desires represented in sentential format. But if cross-modular representations have to be split up into subject-specific components in order to be stored and processed, then it is not clear that content-integration would produce any new cognitive effects that could not have been produced simply by broadcasting the individual components. (This problem reflects the fact that Carruthers differs from many dual-system theorists in not positing dual memory systems. On his view, the development of flexible, cross-modular reasoning did not involve the addition of a new memory system.) This may not be a fatal problem for Carruthers's account, however, and his basic claim that the language system plays a key role in content flexibility is a very attractive one. Here, as in

many other instances, Carruthers's proposals remain interesting and fertile whether one endorses them or not.

It is common to end a positive review by saying that the book reviewed should be read by anyone with a serious interest in the relevant topic. Often this is hyperbole; in this case, it is not. No one will read *The Architecture of the Mind* without being informed, stimulated, challenged, and inspired. It is essential reading.